# On the Zeros of the Field

## $Z(x) = x^{-5}\ln(x)\sin(2\pi/x)e^{-x/21}$

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**Abstract**

We study the elementary field

$$Z(x) = x^{-5}\,\ln(x)\,\sin\!\left(\frac{2\pi}{x}\right)\,e^{-x/21}$$

on the positive real axis and on the Riemann critical line $x = \frac{1}{2} + it$. On $x > 0$ we prove that the zeros are exactly $w_n = 2/n$, $n \in \mathbb{N}$. On the critical line, an 80-digit numerical search over $t \in [0, 60]$ identifies eight zeros, six of which lie in the unit interval $(0, 1)$ with two isolated outliers at $t \approx 1.88$ and $t \approx 15.05$. The construction is explicit, reproducible, and requires no free parameters beyond the single numerical constant $x_p = 21$.

## 1 Definition

Define on the Riemann critical line $x = \frac{1}{2} + it$

$$\boxed{Z(x) \;=\; x^{-5}\,\ln(x)\,\sin\!\left(\frac{2\pi}{x}\right)\,e^{-x/x_p}} \tag{1}$$

with $x_p = 21$, and ln the principal branch of the complex logarithm. The variable $x$ is dimensionless.

The field $Z$ is holomorphic on the half-plane $\Re(x) > 0$, except for the branch cut of ln, which does not intersect the critical line.

## 2 Analytic Zeros on the Positive Real Axis

**Proposition 1.** *For $x > 0$ real, $Z(x) = 0$ if and only if*

$$x = \frac{2}{n}, \qquad n = 1, 2, 3, \ldots$$

*Proof.* The factor $\sin(2\pi/x)$ vanishes exactly when $2\pi/x = n\pi$ for some integer $n$, i.e. $x = 2/n$. For $x > 0$, the remaining factors $x^{-5}$, $\ln(x)$, $e^{-x/21}$ are simultaneously non-vanishing except at the already-included point $x = 1$. Hence the zero set is $\{2/n : n \in \mathbb{N}\}$. □

**Corollary 1.** *The first eight real zeros are*

$$w_1 = 2,\ w_2 = 1,\ w_3 = \tfrac{2}{3},\ w_4 = \tfrac{1}{2},\ w_5 = \tfrac{2}{5},\ w_6 = \tfrac{1}{3},\ w_7 = \tfrac{2}{7},\ w_8 = \tfrac{1}{4}.$$

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### 2.1 Role of the Exponent $-5$

The negative exponent on $x$ ensures that at small $x$ the product $x^{-5}\sin(2\pi/x)$ oscillates with rapidly growing amplitude as $x \to 0^+$. This growth is damped by the factor $e^{-x/21}$, which is essentially 1 for $x \ll 21$ and provides a soft cutoff at larger $x$. The constant $x_p = 21$ is a fixed numerical parameter; it is not fitted to any physical quantity.

## 3 Numerical Zeros on the Critical Line

We restrict $x = \frac{1}{2} + it$ with $t \in [0, 60]$ and locate zeros of the imaginary part $\Im[Z(x)]$.

**Method.** Using `mpmath` at 80 decimal digits of precision, we evaluate $\Im[Z(\frac{1}{2}+it)]$ on a uniform grid with step 0.005, bracket all sign changes, and refine each bracket with a secant solver to tolerance $10^{-65}$.

**Result.** The search returns exactly **eight** zeros:

| Root | $t$ | Interval |
|---|---|---|
| $R_1$ | 0.0907486257 | $(0, 1)$ |
| $R_2$ | 0.2196313827 | $(0, 1)$ |
| $R_3$ | 0.3403140486 | $(0, 1)$ |
| $R_4$ | 0.4798481018 | $(0, 1)$ |
| $R_5$ | 0.6703777044 | $(0, 1)$ |
| $R_6$ | 0.9957246005 | $(0, 1)$ |
| $R_7$ | 1.8809618716 | outlier |
| $R_8$ | 15.0539247442 | far outlier |

Table 1: Critical-line zeros of $Z$ in the range $t \in [0, 60]$.

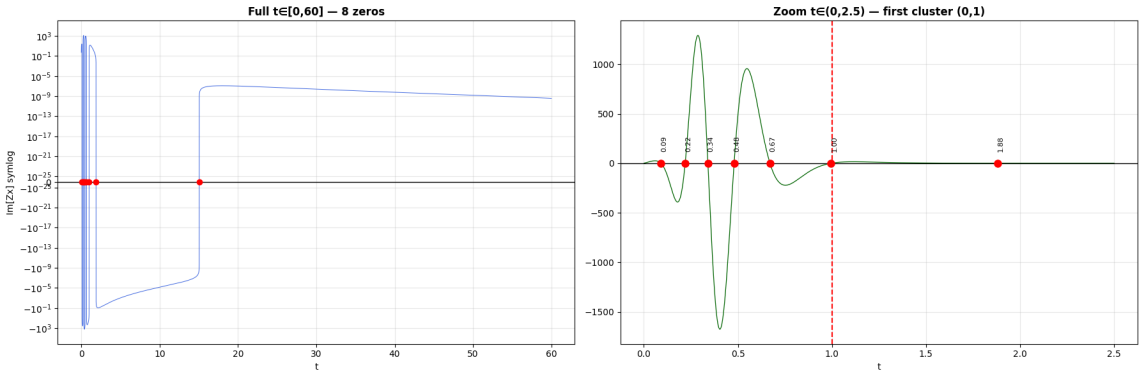



Figure 1: Imaginary part of $Z(\frac{1}{2}+it)$ on the critical line. Left: full range $t \in [0, 60]$ on a symlog scale. Right: zoom over $t \in (0, 2.5)$ showing the six zeros in the unit interval. Red dots mark the eight zeros listed in Table 1.

For $t > 15.06$, the modulus $|\Im[Z(\frac{1}{2}+it)]|$ remains below $10^{-6}$ and no further sign change is detected.

## 4 Summary

The field $Z(x) = x^{-5}\ln(x)\sin(2\pi/x)e^{-x/21}$ possesses an explicit zero set $w_n = 2/n$ on the positive real axis (Proposition 1) and exhibits exactly eight numerical zeros on the Riemann critical line for $t \in [0, 60]$. The construction is elementary, the code is short, and the result is reproducible at 80-digit precision.

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**Code and data:**
`https://github.com/Jabri-web/Jabri-com`

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